



The Bechtel Approach to the Scope and Limits of Mechanistic Explanation

Daniel C. Burnston¹

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Abstract

A huge amount of debate has centered around the “scope and limits” of mechanistic explanation. Critics of mechanism often accuse the mechanistic program of being hegemonic: their strategy is to find genuinely explanatory models that are not (or at least not obviously) mechanistic, and they argue that proponents of mechanism can only incorporate them at the risk of trivializing their thesis. In this paper, I argue that Bechtel’s approach to mechanism is a kind of integrative pluralism, and hence that it is immune from criticisms of this sort. An integrative pluralist thinks that multiple models are required for a given explanandum, but that it is a norm to integrate them in giving explanations. I show how this view avoids both an overly hegemonic approach to mechanistic explanation and criticisms based on limiting mechanism’s scope.

Keywords Bechtel · Dynamical explanation · Explanation · Mechanism · Scientific models · Topological explanation

Introduction

The literature on mechanism is voluminous. Much of it is contentious. And within that contention there is a *lot* of *en passant* conversation. Mechanists and their critics seem, in many cases, to not share the same assumptions, theoretical aims, or philosophical approaches. One might be forgiven for thinking that the debate over the “scope and limits” of mechanistic explanation (Halina 2017) has completely stalled, or maybe even that it got off on the wrong foot in the first place.

Generally, critiques of mechanism proceed by (1) articulating some conditions on mechanistic explanation, (2) arguing that there are some features of complex systems, and/or models describing those features, that fail to meet those conditions but are genuinely explanatory, and (3) concluding that mechanistic explanation is either severely limited in scope or trivial. Mechanists respond in kind, by saying that the models cited in (2) rely, in order to be explanatory, on mechanistic information.

I will argue that Bechtel’s account of mechanism, even as it has evolved over time, has consistently operated outside of this dialectic. This explains why, even as Bechtel’s work has recognized the importance of many different kinds of models and embraced investigation of the many features of complex systems, he has been able to consistently take a mechanistic approach. Far from trivializing the notion of mechanism, I argue, Bechtel’s work shows us why mechanistic investigation and explanation is still central in the investigation of complex systems.

To make the argument, I will pick out and highlight two consistent threads in Bechtel’s thinking about mechanism. Bechtel thinks of mechanisms as *models* and mechanistic investigation as an *explanatory/investigative strategy*. Since all scientific models are partial and all strategies have their limits, it simply falls out of this view that other kinds of models and strategies will be involved in scientific investigation. I will then argue that the best way to read Bechtel’s view is as a kind of integrative pluralism. Pluralism suggests that, particularly for complex systems, multiple kinds of models will be needed for explanation. Integration, however, requires that the distinct models are brought into productive contact in explaining how the system works. If that is the case, then there will be multiple kinds of models and explanatory strategies, *and* mechanism will be one important part of those explanations. This view defuses both

✉ Daniel C. Burnston
dburnsto@tulane.edu

¹ Department of Philosophy, Tulane Brain Institute, Tulane University, New Orleans, LA, USA

overly hegemonic approaches to mechanistic explanation and many criticisms of it.

I proceed as follows. In the next section, “Dialectic,” I further discuss the state of the dialectic between pro- and anti-mechanists. In the section “The Bechtel Approach,” I outline Bechtel’s views on mechanistic explanation as discussed above. In the section “Models and Pluralism: Which Conception?” I explain the difference between two distinct views of pluralism, contrasting integrative with what I call “division of labor” pluralism. In the section “Critiques Revisited,” I show that critics of mechanism adopt division of labor pluralism as an unargued premise, and hence that they do not stand as good criticisms of Bechtel’s approach. Section six, “Advantages,” highlights some advantages of the integrative pluralism approach. The last section concludes.

Dialectic

Meta-Dialectical Issues

In this section, I discuss some of the reasons why mechanists and their critics often seem to be talking past each other. The main issue is what the *commitments* of a mechanistic approach are taken to be. In an attempt to specify their target, critics of mechanism often foist commitments on mechanists that they disavow. This section discusses this state of affairs, as a prelude to (hopefully) moving the dialectic forward.

Criticisms of mechanistic explanation contend that mechanism is either *limited in scope* or *trivial*. If mechanists undertake serious commitments, the reasoning goes, then there will be many examples that do not meet those commitments. These will either be counterexamples if mechanism is construed as a hegemonic approach to explanation (Shapiro 2017), or they will show that only a limited set of explanations in biology are genuinely mechanistic.¹ On the other hand, if mechanists do *not* take on serious commitments, then the view is trivial—it will apply broadly only at the expense of not really being illuminating (for a good exposition of this view, see Silberstein 2021).

I don’t think this way of framing things is right, but some things that mechanists have said have encouraged this way of thinking. For one, some of the earliest philosophical writings on the topic, including both Bechtel and Richardson ([1993]2010), and Machamer et al. (2000), suggested that mechanism is not a *fixed* notion, i.e., that the defining characteristics of “mechanism” can evolve as science progresses.

This definitional hesitancy is well-motivated to some degree because it recognizes, correctly, that scientific concepts and strategies evolve over time, and that is a mistake, metascientifically, to take a snapshot of them and make it definitive. However, it also leaves the intended scope and application conditions for the concept open-ended. If “mechanism” can be adjusted to apply to *any* kind of explanation in the life sciences, then it is indeed trivial, or at least not theoretically deep.

Another move, which is specific to Bechtel and Richardson’s ([1993]2010) treatment, is to adopt a “more-or-less” notion of mechanism. They endorse Simon’s “near-decomposability” as a characterization of mechanism: a system is near-decomposable to the extent that it can be divided into modular subparts. But near-decomposability is a scalar notion, and thus leaves open where to draw the line between systems that are “decomposable enough” to be mechanistic and those that aren’t. As Bechtel and Richardson note, highly integrated systems with feedback tend to resist Simon’s notion of near-decomposability, and this leaves it open to suggest that many or most biological systems that rely on network dynamics are outside the scope of mechanistic explanation.

One can see, in places in the literature, critics of mechanism targeting each of these points. For instance, in a recent paper, Ross and Bassett (2024) target the lack of clear boundaries on the concept. They claim that, without such boundaries, the notion of mechanism is unhelpfully vague, which can lead to both scientific and metascientific confusion. If science employs several notions of “mechanism” with distinct application conditions and commitments, then the concept is not a good guide for how scientists should investigate and explain.

With regards to near-decomposability, Weiskopf (2016) contends that the less we can directly map functions onto distinct parts of a system that operate in isolation, the less we are able to provide mechanistic explanations. He says: “The guiding image of mechanisms [...] strongly suggests that they are made of discrete parts, each of which carries out a dedicated function. [...] The less easily the parts can be separated from one another while retaining their own functions, the further the system drifts away from being mechanistic” (2016, p. 677). This claim directly reflects the idea that mechanistic decomposition is a more-or-less notion, and that many complex systems are likely to be “too far” from ideal decomposability for mechanism to be a useful notion.

So, one worry for mechanism is that the commitments of the view, and its intended scope, are unclear. And without taking on significant commitments, mechanists risk trivializing their view. This is one reason why, when discussing mechanism, its critics tend to first articulate some things

¹ I will only discuss biological explanation here. There is further debate to be had about whether mechanism is a useful concept in, say, economics or social science.

that they think mechanists are committed to. They then provide cases that they think do not exhibit those features, which serve as counterexamples to mechanistic explanation. I will here briefly discuss three of these: *the machine analogy*, *commitment to detail*, and *atomism*. The problem with each of these commitments is that they are not things that mechanists (writ large) endorse or must endorse. This is what leads to the stagnation of the dialectic.

First, the machine analogy. If it is a problem that the application conditions of “mechanism” are too open-ended, one way to fix them would be to center application around a guiding analogy. And since mechanists talk in terms of distinct parts and operations interacting with each other, and since they often cite schematic accounts of causally productive systems as explanatory, perhaps the guiding analogy is to everyday mechanical systems.

However, mechanists have never endorsed a “simple” machine analogy. Bechtel and Richardson ([1993]2010) did express the analogy in *Discovering Complexity*, but they distinguished between “simple” machines such as everyday human devices, and “complex” machines that are open-ended and involve distributed loci of control (cf. Burnston and Tramacere 2023). More recently, Levy and Bechtel (2020) have argued that “mechanism” is a general class, whereas “machine” is a subclass within it. Unlike artifactual mechanisms, they claim, biological mechanisms have heterarchical control structures, where multiple interacting systems constrain each other, instead of being organized as a fixed compositional and control hierarchy (cf. Bechtel and Bich 2021; Bich and Bechtel 2022). Note that while there is a slight shift in terminology here (from distinguishing between simple and complex machines to distinguishing between machines and a broader class of mechanisms), this is not an abandonment of a prior commitment. Similarly, Machamer, Darden, and Craver state that “one should not think of mechanisms as exclusively mechanical” (2000, p. 2, cf. Craver and Darden 2013). Second, critics often contend that mechanists are committed to the idea that more details are better. Mechanists often discuss compositional hierarchies and argue that mechanistic explanations fill in functional roles at successively lower levels. As Ross and Barrett (2024) put it: “The mechanism concept may misleadingly suggest that a causal model contains lower-scale detail or that such details give superior understanding in all contexts” (p. 87; cf. Chirimuuta 2014). However, all mechanists admit that explanations are often abstract and idealized, and that different levels of generality or abstraction can be relevant for different explanatory purposes (Craver and Kaplan 2018; Levy and Bechtel 2013). This undermines the idea that mechanism is always committed to more detailed explanations being better ones. Lastly, the notion of “atomism.” Claiming that mechanism is atomistic is to claim that

it is committed to a particular kind of reductive explanation, where the system must be decomposed into a fixed set of parts with intrinsic functions or operations. Explanation, on atomism, then consists in conjoining the intrinsic functions in the right way to show how they produce the phenomenon of interest. But many mechanists reject atomism, both in its focus on fixed parts with intrinsic functions and its notion of purely bottom-up explanation. Bechtel has argued for years that mechanistic explanation *requires* situating the mechanism in a functional and ecological context (Bechtel, 2009, 2012; Bechtel and Abrahamsen 2009a, b; Wright and Bechtel 2007). On this view, one *cannot* understand the function of individual parts of the mechanism without understanding the context in which the part is operating. Further, (and further rejecting the machine analogy) mechanisms do not have to even be *subsistent* (Bechtel 2020). That is, they can come into being temporarily in response to particular physiological or ecological demands. Burnston (2016a, b, 2021), 2022) has argued at length that functional description in complex systems like the brain is *inherently* contextual. That is, one can mechanistically decompose a complex system, but those decompositions are indexed to behavioral and physiological contexts, and do not have to remain constant across contextual changes.

This section has attempted to diagnose some of the cross talk in the literature between mechanists and their critics. In the remainder of the paper, I will discuss more detailed argument patterns in particular cases. In each case, critics of mechanism discuss a kind of explanation that they take to be *non-mechanistic*—i.e., to not include the kinds of information or description that a mechanistic explanation is committed to giving—and conclude therefore that these examples fall outside the scope of mechanistic explanation.

Criticisms of Mechanism

As I noted in the introduction, critiques of mechanism have a readily recognizable pattern. They consist in describing a certain kind of model and/or feature of a complex system, showing that it is explanatory of some phenomenon, and arguing that it achieves that explanatory status *independently* of whether it cites or describes a mechanism.

I don’t have space to go through every example in detail, so let me say a little bit more about the schematic. Let’s begin with some considerations about “explanation.” Without presuming too much about what explanation consists in (Levy 2025), I think we can agree to the following: all explanations have a *target*. That is, there is some question or phenomenon which the explanation attempts to answer or account for. The *explanans* is a feature or features of the system, described in a model, that answers that question by showing how the phenomenon comes about. I am here

glossing over a lot of questions about the nature of explanation, its relationship to understanding, whether it must cite causes, and so forth, but that is enough to get us going.

Critiques of mechanism argue that, for some given target T , model M , and system feature E , (1) M describes E , (2) E is not a mechanistic feature, and (3) E explains T . We can call this the MET schema, $M \rightarrow E \rightarrow T$. If M does not describe mechanistic features, and E does explain T , then there is nonmechanistic explanation.

Before I list some examples, let me note how much is packed into this simple schema. First, it assumes some more-or-less determinate description of T , such that M/E can stand in the appropriate relation to it. Second, it entails that there is some compelling description of when M is *not* a mechanistic description. Finally, it assumes a notion of explanatory *sufficiency*, such that M describing E is sufficient to explain T , *independently* of other factors. I will return to this in my criticisms below.

I think one can understand most critiques of mechanism according to this schema. The idea is that one can, in a particular case, flesh out the schema by citing some M , E , and T , and arguing that they instantiate the explanatory schema independently of describing a mechanism. And, as noted above, the standard response to this kind of claim is to argue that, in fact, the cited explanation requires, is elliptical for, or is best integrated with a mechanistic description after all.

One can chart this kind of dialectic out in discussion of *dynamical explanations* (Kaplan and Craver 2011; Ross 2015), of *network/topological explanations* (Jones 2018; Matthiessen 2017), of *design explanations* (Dewey 2024; van Eck and Mennes 2017), of *functional explanations* (Picinini and Craver 2011; Shapiro 2017), and so on. Let's just go through a couple of examples.

Topological explanation is explanation that cites classes of organizational features in a system, often described using graph theory. Examples include motif distribution, modularity, scale-invariance, path length, and clustering coefficient. These variables characterize the relationships between nodes in a network at distinct scales; Rathkopf (2018) calls them “pattern variables.” So, T is some system property or behavior, E is a pattern variable, and M is a graph-theoretic description of E . Rathkopf cites Watz and Strogatz's (1998) explanation of disease spreading in terms of the “small world” nature of interaction between carriers.² Huneman (2010) cites scale invariance as explaining the robustness of certain ecological communities against potentially invasive

species.³ Chemero and Silberstein (2013) cite disruptions in network organization as explaining the difference between healthy brains and brains instantiating mental disorders.

Dynamical explanation is explanation that cites continuous patterns of change in a system, described via differential equations and/or state-space representations. So, T is some system property or behavior, M is a state space model, and E is a pattern of change in the system. Philosophers of cognitive science have cited coupling between effectors as explaining certain rhythmic behaviors (Stepp et al. 2011). Ross (2015) cites dynamical models as explaining why many cells with distinct morphologies fall into a few classes of firing patterns. Weber (2008) argues that the patterns of dependency between ion conductances and voltage (as modeled by Hodgkin and Huxley) account for the presence and “shape” of the action potential in a neuron.

I'll discuss some of these examples in more detail below; this initial gloss is just meant to illustrate the schema. To summarize, the anti-mechanists' goal is to limit the scope of mechanistic explanation. To do so, they argue that there is a model M of a system feature E that explains a target T , and that its doing so relies in no way on describing a mechanism. This entails that the MET relation can be described *in isolation* from other—for example, mechanistic—explanations. In the next section, I will describe Bechtel's approach in more detail, and in the subsequent section, I argue that the critiques of mechanism assume a different view of pluralist explanation than Bechtel does.

The Bechtel Approach

In this section, I discuss two important components of Bechtel's approach to mechanism: mechanism as *investigative strategy* and mechanisms as *models*, before moving on to discuss different types of explanatory pluralism.⁴

Mechanism as Investigative Strategy

Going back to *Discovering Complexity*, a significant focus of Bechtel's work has been on the investigative strategies

² A network is “small world” if it is both highly clustered (meaning most connections between nodes are local) and have a short characteristic path length (the number of connections needed to connect any two nodes is low).

³ Scale-invariance means that the degree of connectivity (number of connections per node) is distributed as a power law, with a few nodes having a very high number of connections and most nodes having few.

⁴ I am calling this the “Bechtel approach” to focus on the central themes of his work. But it should be noted that this work has been influenced by many collaborators of Bechtel's, including Adele Abrahamsen, Bob Richardson, Robert McCauley, Arnon Levy, Ben Sheredos, Sara Green, Ingo Brigandt, Linus Huang, Jason Winning, Leonardo Bich, and Andrew Bollhagen.

suggested by mechanistic analysis.⁵ Within that focus, mechanistically decomposing a system is *one* important strategy, but not necessarily the only one. In *Discovering Complexity*, the focus was on the twin strategies of decomposition and localization. Often, investigation begins with “direct localization,” in which a complex phenomenon is posited to occur fully in some part of the system. Examples include the purported localization of language production in Broca’s area (Brodman area 45) or vision in the primary visual cortex. Inevitably, in complex systems, these simple localizations will prove insufficient. The failure of direct localization spurs further investigation, which both contributes to decomposing the phenomenon into distinct aspects and to the more complicated localization of those aspects in distinct parts of the system.

Bechtel’s characterization of decomposition and localization has evolved and been expanded since *Discovering Complexity*. In his shared work with Robert McCauley (McCauley and Bechtel 2001), they posit that the relationship between psychological and neuroscientific kinds is best understood in terms of “heuristic identity theory.” On this view, a provisional identification between a functional or experiential property is posited as a defeasible guide to investigation in the brain. Similarly, in a series of works in the mid-2000s, Bechtel posited that decomposition and localization are *compatible with* and *equally necessary to* strategies that contextualize the parts of a mechanism in terms of their contribution to functions at higher levels of the system, as well as the environment in which the mechanism operates. That is, as the title of the paper states, “Mechanists must be holists too!” (Bechtel 2016).

This is where a slight, but important, modification comes in. It is not as though for certain kinds of complex systems, one “exits the mechanistic program” (Bechtel and Richardson [1993]2010, p. 116). Rather, investigation of a *single* complex system involves multiple kinds of strategies, some of which are based explicitly on localization and decomposition and some of which aren’t. Critically, these methods benefit from each other. We understand the parts better by understanding how they are situated in the whole, and we understand the whole better by understanding how the parts contribute to its behavior.

This part of Bechtel’s approach has been expanded out in recent years. For instance, network analyses start by analyzing a system’s overall organization, rather than by first localizing responsibility for a phenomenon in some part of the mechanism. This view of organization constrains

the search for functional mechanisms within the system (Bechtel 2017b, 2019). Similarly, rather than thinking about mechanisms as purely passive systems, one should think about them as part of endogenously active systems, hence situating the operation of parts within a larger dynamical context (Bechtel 2012). Moreover, one should think of mechanisms in relation to the control and direction of free energy to perform work—mechanisms, that is, should be viewed partially in terms of the constraints they put on the distribution of free energy within the system (Winning and Bechtel 2018). And finally, one should think of those constraints as contributing to the maintenance of autonomy in a biological agent, where the work done by the system is meant to maintain its own organization (Bechtel and Bich 2023; Bich and Bechtel 2022).

In summary, the investigative strategy advocated by Bechtel as partially constitutive of the mechanistic approach has never been hegemonic. It has consistently been pitched in terms of *interacting* and *mutually supporting* strategies. That is, while each strategy is partial and could be pursued independently (and may be at certain stages), the strategies benefit from interaction. Thinking about broader organizational properties shapes the way we think of the parts of a mechanism. And thinking about parts of the mechanism contributes to understanding how those organizations maintain the system.

Mechanistic Descriptions as Models

Bechtel and Abrahamsen (2005), in their seminal paper on mechanistic explanation, propose an *epistemic* account. This means that explanations comprise cognitive activities of scientists. One way to cash out this claim is to do so in terms of *models* of the world that help scientists understand the phenomenon. Mechanistic models can have a number of vehicles. While they can be described verbally, they are more often portrayed as *diagrams*. They also can be represented mentally by the scientist. Often, these “internal and external” representations are combined, and a scientist mentally simulates the operation of the mechanism portrayed in the diagram in order to understand how the phenomenon comes about.

The notion of mechanistic descriptions as models is borne out in several further developments of Bechtel’s work. First, Bechtel has focused on the cognitive role that diagrammatic representations of mechanisms play in the cognitive processes scientists undertake, and second, he has focused on the way that mechanistic descriptions are idealized. In his work with the Working Group on Diagrams in Science (<https://mechanism.ucsd.edu/bill/WORGODS/index.html>), the group focused on diagrams as drivers of epistemic practices. Sheredos et al. (2013) contend that

⁵ Of course, Bechtel is not the only mechanist to discuss this. Craver and Darden (Craver and Darden 2013) give an extensive discussion of the kinds of investigation that are constitutive of the search for mechanisms. Levy (2012) distinguishes metaphysical, explanatory, and investigative approaches to mechanism.

diagrams *constrain and afford* hypotheses about the system. Burnston et al. (2014) provided a case study that showed that views of the organization of a mechanism can coevolve with the development of novel ways of diagramming the mechanism. Sheredos and Bechtel (2016) further argue that diagrams are vital tools for scientific *imagination* (cf. Sheredos 2017). As such, mechanistic models guide the search for new hypotheses about the system.

Another aspect of this line of thought has been to note the many ways in which mechanistic descriptions are idealized. In an important paper, Bechtel (2015a, b) notes that many biological systems do not have a well-defined spatial or temporal scale. If that is the case, then the idea that mechanisms are systems with fixed boundaries and clear constitution relationships between levels cannot be maintained. This does not, however, undermine the importance of *imposing* structure on the system. Even if the spatial and temporal boundaries that we represent the system as having do not map onto firm distinctions in nature, imposing them is necessary for thinking about the way the system works. Idealization is arguably a definitive feature of scientific models—so, if descriptions of mechanisms are regularly (or intrinsically) idealizations, then this supports viewing mechanistic descriptions as models. Of course, the fact that models of mechanisms are idealizations does not *impugn* them in any way, because it is a feature they share with all other types of models.

Similarly, and partially because of the particular idealizations they impose, all models have *limits*. That is, models are partial and best used for certain purposes. Here we can see a commonality between models and strategies. And as with strategies, Bechtel has consistently argued that different kinds of models are best seen as complementary rather than competing. In particular, explaining what a system does requires, in most cases, multiple models, and the relationships between models can vary as the pragmatic aim of the scientist does. In an early statement of this view, Bechtel and Abrahamsen (2010, 2012) argue compellingly that understanding circadian systems requires both mechanistic and dynamical descriptions, and they offer this as a normative framework for cognitive (neuro)science. Each kind of model has its important role; mechanistic models are used to identify the parts that play a role in the system and their organization, and the dynamical description helps to re-compose the phenomenon from the organization of the parts.

Again, this general perspective has played an important role in the further development of Bechtel's approach. Bechtel (2015a, b) and Levy and Bechtel (2013) have argued that network representations are helpful both in abstracting and generalizing mechanistic explanations. Modeling the system as a network makes abstract organizational properties salient in a way that a more detailed mechanistic

description does not (cf. Green et al. 2018). Another example involves the representation of quantitative relations. In accounting for fine-grained phenomena, such as particular temporal patterns in circadian mechanisms, it is necessary to represent relationships between quantities in the mechanism, information that is not apparent in, for instance, a diagram of a mechanism (Bechtel 2017a; Bechtel and Levy 2020; cf. Burnston 2016c, 2017).

As with strategies, the picture that emerges here is one on which multiple *interacting* models are important for understanding how the system works. Each model has its advantages for representing the system, but understanding a particular phenomenon requires multiple of them. In sum, Bechtel's view of models, like his view of strategies, is *neither hegemonic nor disjoint*. No particular kind of model “owns” explanation, and many different kinds of models can contribute to explanation, even of the same target. No given model is “elliptical” for another—each makes a positive explanatory contribution in virtue of the way it represents the system. But the flip side of this is that individual models are unlikely to be explanatory “purely on their own.” Again, the picture we get is of the interaction of multiple kinds of description rather than a dissociation between them.

Models and Pluralism: Which Conception?

The “Bechtel approach” that I articulated above stresses the multiplicity of explanatory models and strategies but sees multiplicity as requiring interaction between distinct types. If that is the case, then simply citing that there are other models involved in explanation is no criticism of mechanism per se. In this section, I explain how critics of mechanism and the Bechtel approach assume different notions of explanatory pluralism. My aim in this section is diagnostic: critics of mechanism do not explicitly endorse the positions I am attributing to them. I think they *assume* something like these positions, because these positions explain why they take their arguments to be compelling critiques of mechanism. Rejecting these positions, then, renders their arguments against mechanism unconvincing.

Let's go back to the MET schema:

$$M \rightarrow E \rightarrow T.$$

This is the relation between a model, an explanans, and a target. As noted above, I think critics of mechanism make substantive assumptions about the nature of this relation, as it pertains both to models and to pluralism, that is simply different from the Bechtel approach. Speaking broadly, let's distinguish two kinds of accounts of models. On *mapping* accounts, a model represents a system via some mapping between the features of the model and the features of the system. Standard options for the mapping relation include

similarity and isomorphism (for discussion, see Frigg and Nguyen 2020; Weisberg 2013). Glossing over the specifics, the assumption is that the model can be decomposed into some set of model features $F_1 \dots F_n$, and these correspond to the system features $E_1 \dots E_n$ being represented. On *inferentialist* accounts, models allow an informed user to make inferences about the system of interest (Khalifa et al. 2022; Suárez 2004a, 2024b). While there is disagreement about how reference is achieved in inferentialist accounts, all inferentialist accounts deny that some mapping relation is required for the model to represent the system. Speaking broadly, let's distinguish two kinds of pluralism. On what I call "division-of-labor" pluralism (Burnston 2019, 2024), the system of interest is divided into distinct features, and the phenomenon of interest is divided into distinct explananda. The latter are the potential targets for explanation T, and the former the potential explanantia E. Division of labor pluralism assumes that there are distinct kinds of models and/or investigative strategies that are appropriate to distinct subsets of E, and further that distinct subsets of T are appropriately explained by those subsets of E. The outcome of this is a "parcellation" each of T, E, and M into subsets, where the intersection of subsets is minimal. That is, the phenomenon and the system are both divided into a series of $E \rightarrow T$ relations, with distinct models applying to distinct subsets of those relations (see, e.g., Potochnik 2017).

In "integrative" forms of pluralism, the parcellation does not work this way. Rather, the relationship between features of T, system features E, and model types M is many-many. Most frequently associated with Mitchell (2003, 2020), this view suggests that *for a given target* T, multiple distinct types of models (and system features described by them) will be required to account for T (cf. Hochstein 2017). That is, the parcellation of T *does not* correspond directly to the parcellations of M and E. Instead, multiple Ms and Es contribute, in an additive fashion, to the explanation of a *given* T. Bechtel's early writing on integration primarily discussed integration between distinct disciplines or fields, but Bechtel and McCauley (2001) cite "explanatory connections" between different kinds of descriptions that lead to "fruitful interaction," beyond just "mere correlation" of different descriptions. I read this as signing on for the notion of additive explanatory benefit to integration.

There are significant questions about what this "additive" aspect consists in—i.e., what exactly it is about combining models that is better, explanatorily, than simply using them separately. I don't have space to pursue this in depth here, so instead will just note that several theorists have tried to clarify a notion of "explanatory relevance," in which one kind of theory or model is thought to contribute to explanations given by another (Bolinska and Sali 2025; Nathan 2017). Nathan helpfully describes this as follows: A and B

are explanatorily relevant to each other "when conceptual advancements and testable results in A are necessary for raising explananda and providing explanantia in B, and vice versa" (2017, p. 176). I will adopt this definition for characterizing the "additive" notion of integration throughout.⁶

Basically, what I contend is that critics of mechanism assume mapping and division of labor accounts, whereas Bechtel assumes inferentialist and integrative accounts. While Bechtel does not use these terms himself, I contend that this package of views makes good sense of his general approach and, further, defuses many of the criticisms of mechanism. Let me begin by emphasizing how the mapping/division-of-labor views would encourage the criticisms of mechanism. On the schema above, each aspect of the phenomenon, symbolized as distinct Ts, has its proper and correct kind of feature/model that explains it. The critic of mechanism construes mechanists as contending that *all* Ts must be explained purely by mechanistic models. Their arguments against mechanists are based on distinguishing some Ts, and some subset of Es, where the model explaining that particular $E \rightarrow T$ relation does not cite a mechanism. On the mapping view of models, this is the suggestion that there are disjoint sets of features/explananda in the system, that disjoint types of models represent. On the division-of-labor view, the phenomenon is divided up into distinct aspects, each with its appropriate kind of model explaining it.

The inferentialist/integrative view simply denies that the set of $M \rightarrow E \rightarrow T$ relations can or must be parsed in this way. Inferentialism, in not positing a discrete mapping from models onto system components, opens the way for distinct inferences, derived from distinct models, to contribute additively to an understanding of each given T. Integrative views propose it as a norm that we understand a given T better when we bring multiple modeling resources to bear on it. Note here, and this is a key point: Neither view contends that *every particular* investigation of an $E \rightarrow T$ relationship must bring these resources to bear. What they suggest is that understanding each given T should, in the long run, rely on multiple distinct models.

Is this a good way to make sense of Bechtel's approach? I contend that it is. Recall that, both in terms of investigative strategies and models, Bechtel stresses that mechanistic models/investigations are *one part* of explanation. In complex systems, most Ts will require integrating information from multiple sources, discovered through different strategies and expressed through different models. In sum, *individual models/strategies are not explanatorily sufficient on*

⁶ Nathan is officially discussing "unification" rather than integration, but as Bolinska and Sali note, the notion of explanatory relevance can be extended to explanation with distinct models as well. Other discussions of integration can be found in Brigandt (2010, 2013) and O'Malley (2012).

their own. What it means to be a mechanist, on this view, is to think of mechanism as a core component in a set of modeling/investigative strategies that are brought together to the understanding of phenomena. Our understanding of a system, and each T associated with it, is better to the extent that we can explain how distinct parts and their organization within the system contribute to T. What it does *not* mean to be a mechanist, is to say that only mechanistic models can contribute to explanation, that all non-mechanistic models must turn out to be elliptical for mechanisms after all, or anything like that.

From the inferentialist/integrative perspective, what critics of mechanism do is half right. If explanation requires multiple models, as all pluralists agree, we should expect different kinds of models to make different contributions to explanations. And it is good philosophical work to distinguish the different explanatory contributions made by different kinds of models. It is a mistake, however, to infer from this that explanation with these models proceeds independently of mechanistic modeling. In the next section, I discuss some of the extant criticisms of mechanism in light of this framework.

Critiques Revisited

Let's look at a couple of examples of critiques in more detail. To do so, we can add slightly to our schema. In each of the following arguments, one of the considerations given is the particular role of *abstraction* in the explanation. So, a model M citing explanandum E, while *abstracting away* from mechanistic factors A, explains target T. Let's call this the 'MEAT' schema. Each criticism attempts to show that the system features E and models M *are* explanations of particular kinds of phenomena T. Since they are explanations of T, and intentionally leave out mechanistic information (A), then they are non-mechanistic explanations. As noted above, this involves notions of both explanatory sufficiency for T and independence of the explanation from the features that are abstracted away (A).

Huneman (2010) argues that robustness to invasive species is determined by the network structure of interactions between species in a population. Importantly, he argues that these are *not* particular types of causes. It doesn't matter whether an interaction is a predation relationship, competition for resources, and so on. The thing that matters is the distribution of the connections. If the connections are distributed according to a power law (which is what makes the network scale-free), then the network shows robustness. This is because most interactions with an invasive species will occur within a smaller network and not affect the

smaller number of network "hubs" that have higher levels of connectivity.⁷

Jones (2014) focuses on the opposite of robustness, fragility. In particular, the human immune system is extremely susceptible to severe interruption after infection by HIV. The reason for this is the topological structure in which HIV's host cells, CD4+T cells, play a part. CD4+T cells occur at the center of a "bowtie" structure. This means that many pathways activated by distinct stimuli converge on the CD4+ cells, and that the CD4+ cells subsequently interact with many downstream processes. Due to the centrality of the CD4+ cells in this structure, the immune system is highly vulnerable in cases of CD4+ cell interruption, which is what HIV causes.

Ross (2015) cites the explanation of distinct "classes" of neurons: Class I neurons, which increase their firing nearly linearly with increased input, and Class II neurons, which begin to fire only at a high degree of input and do not vary much after that. Ross describes how low-dimensional models of neural activity, which conceptualize activity as only based on an "excitation variable" and a "recovery variable," explain the Class I pattern of firing. In particular, these models exhibit a bifurcation between a quiescent state and a firing state that begins at a certain value of input and continues to increase as that input increases. Importantly, *any* neuron that exhibits this bifurcation will fire in a Class I pattern, independently of its morphological details.

Weber (2008) notes that the conductance model given by Hodgkin and Huxley shows *how* a certain pattern of dependence between conductivity and voltage for a set of charged ions could produce an action potential. Further, he claims that the correct fitting of the variables in the conductance model captures the particular *shape* of the action potential—i.e., its evolution from threshold potential to peak, its subsequent hyperpolarization, and its refractory period. Weber notes that it would be possible to be *wrong*, as Hodgkin and Huxley arguably were, about the mechanisms by which those dependencies are instantiated, without undermining the importance of the conductance relationships for explaining the shape of the action potential.

We can summarize each of these arguments in terms of our MEAT schema. The schematics describe the style of explanation offered, and the quotes show that each theorist thinks the model they cite *is* an explanation of the target, *independently* of mechanistic information.

⁷ I should note that, in other work, the theorists cited here do sometimes talk about closer integration between mechanism and other forms of explanation—see, e.g., Huneman (2018), Ross (2018), and Jones (2018). I am critiquing the kinds of arguments given in these instances, which "emphasize the contrast between[...] types," of explanation (Huneman 2010, p. 225) and claim that non-mechanistic models are "explanations per se" (2010, p. 226).

Huneman:

- Graphical model (M) → Scale-free property (E) → Types of interaction (A) → Robustness to invasion (T). (2010, p. 213)
- “Besides mechanistic explanations, there is a kind of explanation that relies upon “topological” properties of systems in order to derive the explanandum as a consequence, and which does not consider mechanisms.” (2010, p. 214)
- “In some cases, the explanandum will be wholly explained by S [the system] having T_i [topological features].” (2010, p. 218)

Jones:

- Graphical model (M) → Bowtie structure (E) → Specifics of pathways (A) → Vulnerability to CD4+ cell depletion.
- “Rather than accounting for their explananda by showing them to be outcomes of the operation of a mechanism or by subsuming them under biological laws, topological explanations show their explananda to be consequences of a system’s topological properties.” (2014, p. 1135–36)
- The topological model “is not associated with [...] mechanistic explanation, because [the] explanation does not show how the parts of a mechanism operate upon each other to produce a particular effect. The [...] explanation, instead, appeals to topological properties.” (2014, p. 1150)

Weber:

- Hodgkin Huxley model (M) → Conductance/voltage dependence relation (E) → Causal interactions underlying dependence relations (A) → Shape of the action potential (T).
- “Mechanisms in the narrow sense much discussed in recent philosophy of biology and neuroscience [...] are by no means necessary for giving causal explanations of natural phenomena.” (2008, p. 1006)⁸

Ross:

- Reduced dimensional spiking model (M) → Bifurcation in state space (E) → “Microstructural details” differentiating cell types (A) → Class I excitability (T).

- “The techniques of dynamical systems neuroscience are used to explain why such universal behaviors are exhibited by physically distinct systems, as opposed to just providing descriptions or predictions of these behaviors or revealing their underlying causal mechanisms.” (2015, p. 51)
- The models “provide [...] an explanation, despite failing to meet [...] mechanist requirements.”

What is interesting about these examples, from the integrative pluralistic perspective I outlined above, is that *all of $M \rightarrow E \rightarrow A \rightarrow T$ are exactly what we should expect*. The integrative pluralist view says that multiple models are required for explanation and that all models abstract and idealize, some in very different ways. The fact that these models are required to explain, and that they make a distinctive contribution to doing so, is exactly what the integrative pluralist suggests.

Here is the main difference. For the integrative pluralist, the fact that there *is* a distinctive explanatory relationship between $M \rightarrow T$, and even the fact that that relation is *necessary* for explaining T, simply doesn’t entail that *only* M is relevant for explaining T. The only thing that justifies the quotes given after each instance of the schema is division-of-labor pluralism, which thus acts as an unstated premise in each argument. If I am right that the Bechtel approach to mechanisms is committed to integrative pluralism, then the Bechtel approach simply rejects this premise and says that the argument is unsound.

All I have argued so far is that the critiques given of mechanism are not critiques against Bechtel’s version of mechanism. Critics of mechanism make an assumption that Bechtel rejects. The question then is, which is the overall better approach: the division-of-labor one or the integrative one? While I can’t hope to fully settle this in this paper, in the next section, I will make some broad suggestions in favor of the Bechtel approach.

Advantages

Let’s start back with the notion of explanatory relevance. As noted, integrative pluralism, construed in terms of explanatory relevance, suggests that multiple distinct kinds of models, describing multiple distinct aspects of the system, are relevant for each explanatory target. What does this mean? For sake of discussion, I’ll assume that explanation involves the ability to answer “what if things had been different” questions about the target. On this view, integrative pluralism suggests that for each T there will be multiple M that allow us to answer these kinds of questions.

⁸ Weber also says that the model has “explanatory status,” and cites Hodgkin and Huxley as saying that the model is explanatorily “sufficient” for certain purposes despite needing “no correct physical interpretation” (2008, p.1003).

The division-of-labor view denies explanatory relevance of mechanistic models for each chosen T. This is how we should interpret the oppositional claims given above—i.e., that it is a certain M and *not* other Ms that explains T. The way to assess the question about explanatory relevance is to ask if mechanistic models provide deeper understanding of these phenomena, *in addition* to the critic's chosen M, in terms of answering a broader range of questions.

From the integrative pluralist perspective, the answer seems to be an obvious yes, and this illustrates a very basic point. The fact that a nonmechanistic model is an abstraction over certain mechanistic details does not mean that it is an abstraction over *just any* mechanistic details. That is, just as mechanists have always said, there is a fact of the matter about what kinds of mechanistic organizations can produce the E described by M. Knowing what those organizations are, and what determines membership in that kind of class, *is all relevant* to understanding T. Put simply, if the system did not have the right kind of mechanistic organization, then the kinds of Es described in M would not occur. Critics of mechanism assume that abstraction over mechanistic facts renders those facts explanatorily irrelevant. But it doesn't.

We can go through and make this point about the examples. By very slightly shifting the explanatory question or by considering the system more broadly, we can easily see how mechanistic information is relevant to understanding these systems. (I will keep these considerations general here; for more details in specific cases, see Burnston 2021, 2022; 2025)

Ross (2015) claims that morphological differences are irrelevant in explaining the distinction between Class I and Class II neurons. But, given the low-dimensional model, we can of course ask what kinds of morphological/organizational structures are required to be part of one class versus another. Do they require certain dendritic distributions? Certain receptor types? Is there any way we could intervene (genetically, optogenetically, or otherwise) to *make* a Class I neuron into a Class II neuron? The fact that this information is not employed in describing the bifurcation point specifically does not mean that it is irrelevant to explaining different classes of neurons and how they behave.

Huneman (2010) claims that the type of interactions between species is irrelevant to the explanation of robustness. And it is certainly impressive that one can abstract over this information in accounting for differences between more and less robust networks. But of course, one can't just abstract over *any kind* of relation. A particularly virulent pathogen, for instance, might start out in a sequestered population but be able to "bridge" levels. If it does, then the reason why robustness will break down in this case will depend on the properties of the pathogen, the way it interacts with its hosts, etc. This entails that the abstraction and

the robustness only apply within a *range* of biological relationships. And mechanistic information is relevant to understanding that range and its boundaries.

Jones (2014) claims that the place of CD4+ cells in the bowtie structure renders the system particularly susceptible to CD4+ cell depletion. But of course, every link in the topological structure is a mechanistic link—the interactions between cells in the topological model are mediated by molecular signaling, binding, affinity, and so on. The topological description is indeed a useful abstraction that allows us to understand the vulnerability of the system better, but the *reason* that topology is instantiated is because of the mechanistic organization of the system. And intervening on the system will involve intervening on those mechanistic interactions.

Weber (2008) claims that patterns of dependency explain independently of mechanistic details. But the fact that one can distinguish a model describing dependencies from a model providing mechanistic organization simply doesn't imply (either historically or in terms of our current understanding) the explanatory independence of one vis-à-vis the other. The fact that the conductance model *does* provide distinct and explanatorily relevant information about the shape of the action potential *does not* mean that it does so at the expense of understanding how the mechanism works.

These considerations show that the isolation of a specific $E \rightarrow T$ relation in the cases is relatively artificial. Critics of mechanism conflate a *correct* view that different kinds of models provide nonredundant explanatory information with the *incorrect* view that explanation by one type of model rules out explanation by another. The fact that some information is backgrounded, abstracted over, or even unknown, in a particular case doesn't mean it's explanatorily irrelevant in that case.

Put another way, critics of mechanism often take a "snapshot" of a particular instance of explanatory work, and think that it stands alone, independently of other knowledge of the system. There is no reason to think this. Integrative pluralism, in my view, correctly captures the fact that at different times in investigation, different models come to the fore to fill gaps in our understanding. But doing this doesn't *obviate* the rest of our understanding of the system, nor render the things we have already learned explanatorily otiose, nor mitigate against the idea that we should deepen our understanding by bringing in other forms of information.

This view, I contend, further captures a bunch of interesting facts about models. For one, models have many uses, and those uses have many relations to each other. Sometimes a model is used to fill in a gap in our understanding. Sometimes it is used to help conceptualize data or to suggest experimental interventions. Sometimes it is an abstract rendering that points out where details need to be filled

in. Sometimes it intentionally abstracts from those details to help isolate a broader pattern. In one instance, a model might be used to help explain a particular phenomenon, in another, it might be used to constrain the search space on hypotheses of another type. *No one kind of model plays only one kind of role.* Explaining a phenomenon involves bringing the right kind(s) of information (as described in a model) to the target, in the right combination for the kind of question being asked. As the integrative pluralist suggests, target types and model types are not related one-to-one.

So, finally, let me end with the trivialization worry. As noted in the second section, the trivialization claim is often motivated by saddling mechanists with claims that they in fact reject. But mechanists do make a claim both about how we investigate and how we understand complex biological systems. They claim that a productive strategy for understanding a biological system is to investigate its mechanistic organization. They also claim that we understand the system better when we understand the ways in which its component parts are organized to produce phenomena. This pragmatic aspect has always been part of the mechanistic program and expresses norms of both investigation and explanation.

If “mechanism” meant “explanatory hegemony of mechanistic models,” then the mechanism thesis about explanation would be wrong. But Bechtel, on my reading, has never meant that. The Bechtel approach to mechanism, as I understand it, is simply immune from many of the criticisms of mechanistic explanation and avoids them without trivializing the mechanism program.

Conclusion

I have attempted to bring some clarity to the discussion surrounding the “scope and limits” of mechanistic explanation. I have characterized Bechtel’s view of mechanism as a kind of integrative pluralism and argued that most critiques of mechanism do not speak against this approach. If I am right, we should recognize that an equally important counterpoint to differentiating distinct kinds of explanatory models is showing how they mutually contribute to explanation. This, I suggest, has always been a part of Bechtel’s program, and stands still as a good guide in philosophy of science.

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